MILITARY SPECIFICATION

SWEATER, MAN'S (FLAME RETARDANT)

This specification is approved for use by the Navy Clothing and Textile Research Facility, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a knitted pullover man's sweater, made of a modacrylic/wool blend.

1.2 Classification. The sweater shall be of one type, and shall be procured in the following sizes as specified (see 6.2):

Schedule of Sizes

Extra-Small	Small	Medium	Large	Extra-Large
(32-34)	(36-38)	(40-42)	(44-46)	(48-50)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760-2490 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

V-T-276	- Thread, Cotton
NN-P-71	- Pallet Material Handling, Wood, Stringer Construction, 2 Way and 4 Way
DDD-L-20	- Label; for Clothing, Equipage, and Tentage (General Use)

MILITARY

MIL-P-15011	- Pallet, Material Handling, Wood, Post Construction, 4 Way Entry
MIL-B-17757	- Boxes, Shipping, Fiberboard (Modular Sizes)
MIL-T-43548	- Thread, Polyester, Cotton Covered and Rayon Covered
MIL-C-43665	- Cloth, Wool: Mothproofing Treatment of

STANDARDS

FEDERAL

FED-STD-191	- Textile Test Methods
FED-STD-751	- Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-147	- Palletized Unit Loads
MIL-STD-1491	- Glossary of Knitting Imperfections

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

LAWS AND REGULATIONS

US POSTAL SERVICE MANUAL

(Copies of the manual may be obtained from the Superintendent Of Documents, US Government Printing Office, Washington, DC 20402.)

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2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and supplement thereto, if applicable.

THE COLOR ASSOCIATION OF THE UNITED STATES, INC.

Standard Shades for Threads - 1968

(Applications for copies of color cards should be addressed to the Color Association of the U.S., 24 E. 38th Street, New York, NY 10016.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Applications for copies should be addressed to the American Trucking Association, ATTN: Traffic Department, 1616 P Street, N.W., Washington, DC 20036).

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Applications for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606).

US DEPARTMENT OF AGRICULTURE

Methods of Test for Grades of Wool Top

(Applications for copies should be addressed to the US Department of Agriculture, Agricultural Marketing Service, Washington, DC 20402.)

THE AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS
AATCC TM 135-1978

(Applications for copies should be addressed to The American Association of Textile Chemists and Colorists, P.O. Box 12215, Research Triangle Park, NC 27709).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Guide sample. Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variations from this specification may appear in the sample in which case this specification shall govern.

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3.2 Standard sample. The finished sweater shall match the standard sample for shade, and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced.

3.3 First article. When specified, the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3 and 6.2).

3.4 Material.

3.4.1 Fiber identification.

3.4.1.1 Modacrylic. The modacrylic fibers shall be 3 to 5 inch variable cut staple length with an irregular, off-round cross-section (see 4.4.1).

3.4.1.2 Wool. The wool fibers shall be from fleece or pulled sheep's wool, or a combination of both, not lower in grade than 54's U.S Standard. The use of reprocessed, reused, or waste wool is prohibited. The wool shall be mothproofed in accordance with MIL-C-43665 (see 4.4.1). The wool grade shall be tested in accordance with 4.4.1.1.

3.4.2 Yarn. The yarn for knitting and for the cover sewing (see Table III, 1/) shall be a blend of 70% (± 5%) modacrylic with the remaining percentage wool when tested as specified in 4.5.

3.4.3 Knitted fabrics. The knitted fabrics for the component parts of the sweater, constructed as specified in 3.4.3.1 through 3.4.3.3, shall conform to the physical requirements shown in Table I when tested as specified in 4.5.

Table I - Physical requirements of knitted component parts

Component	Stitch Type	Wales per inch (face)(Minimum)	Courses per inch (Minimum)
Body	Half cardigan	7	15
Skirt	1X1 rib	12	20
Sleeves	Half cardigan	7	15
Wristlets	1X1 rib	12	20
Collarette	1X1 rib	12	20

3.4.3.1 Body and skirt. The body and skirt shall be continuously knitted together to applicable body width (circular knit without side seams) or with two full side seams (see Table III, operation 4). The skirt shall finish not less then 3 1/4 inches nor more then 3 3/4 inches in height (wale direction).

3.4.3.2 Sleeves and wristlets. The sleeves and wristlets shall be continuously knitted together. The sleeves shall be shaped as shown in Figure 1 and conform to the measurements specified in Table IV. The wristlets shall finish not less then 3 1/4 inches nor more then 3 3/4 inches in height (wale direction).

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3.4.3.3 Collarette. The collarette shall be of a circular knit seamless tubing or flat knit in one piece, to finish as specified in Table III, operation 6. The collarette shall finish not less than 2 inches nor more than 2-1/2 inches in height (wale direction).

3.4.4 Thread. The thread for seaming and stitching the sweater shall be cotton, conforming to type IC2, ticket no. A, 3-ply, of V-T-276. As an alternate, the thread may be covered polyester, conforming to MIL-T-43548, ticket no. 50. The thread shall be dyed Black AA, Cable No. 66043 or Blue AB, Cable No. 66044.

3.4.4.1 Colorfastness. The dyed thread shall show fastness to light, laundering (after 3 cycles), wet-dry cleaning, and perspiration equal to or better than the standard sample. When a standard sample is not available, the dyed thread shall show a minimum of "good" fastness to light, laundering (after 3 cycles), wet-dry cleaning, and perspiration. Testing shall be performed as specified in V-T-276 or MIL-T-43548, as applicable.

3.4.5 Label. Each sweater shall have a combination identification, instruction and personal label conforming to Type VI, classes 10 and 15 of DDD-L-20. The marking shall show colorfastness to laundering and wet-dry cleaning. The inscription and format shall be as follows:

NAME:
SERVICE NO:
SWEATER, MAN'S (FLAME RETARDANT)
FIBER CONTENT: 70% MODACRYLIC/ 30% WOOL
CONTRACT NO.: DLA 100-00-0-0000 (Example)
STOCK NO: 8405-00-000-0000 (Example)
SIZE: MEDIUM (40-42) (EXAMPLE)
NAME OF CONTRACTOR:

LAUNDERING INSTRUCTIONS

Machine wash, cold water, knit or delicate cycle. Tumble dry warm. Remove promptly. Do not use chlorine bleach.

OR

Use NAVEDTRA 414-01-45-81, Formula III

3.5 Color. The color of the finished sweater shall match Blue Shade 3346.

3.5.1 Matching. The color shall match the standard sample under artificial daylight having a color temperature of 7000 (± 500) Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2850 (±100) Kelvin.

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3.5.2 Colorfastness. The dyed finished sweater shall show fastness to light (40 SFH), laundering, wet-dry cleaning, perspiration, and crocking equal to or better than the standard sample when tested as specified in 4.5. When a standard sample is not available, the dyed finished sweater shall show a minimum of "good" fastness to light (40 SFH), perspiration, and crocking, and a minimum of "fair" fastness to laundering and wet dry cleaning when tested as specified in 4.5.

3.6 Nonfibrous material content. The starch and protein content, including chloroform-soluble and water-soluble material of the finished dyed sweater, shall not exceed 3.0 percent when tested as specified in 4.5.

3.7 Dimensional stability. The finished sweater shall not shrink or elongate more than 7% in either the length (wale direction) or width (course direction) after being subjected to five washing and drying cycles. Testing shall be performed as specified in 4.5.

3.8 Bursting strength. The finished sweater shall show a minimum bursting strength of 80 pounds when tested as specified in 4.5.

3.9 Flame resistance. The finished sweater shall have an average time of after flame of not more than 1 second, not more than 3 seconds average after-glow time and an average length of char of not more than 2 inches, when tested as specified in 4.5.

3.10 pH. The pH value of the water extract of the finished dyed sweater shall be not less then 4.0 nor more than 8.0 when tested as specified in 4.5.

3.11 Weights. The weights of the finished sweaters shall be in accordance with Table II when tested as specified in 4.5.

Table II - Finished weights of sweaters

Wgt/dozen lbs. (minimum) conditioned	Extra-Small (32-34)	Small (36-38)	Medium (40-42)	Large (44-46)	Extra-Large (48-50)
	8 lbs. 4 ozs.	9 lbs. 8 ozs.	10 lbs. 8 ozs.	12 lbs.	14 lbs.

3.12 Design. The sweater shall be of a pullover style, with or without side seams, with a one piece mock-turtle neck (crew) collarette, rib knit skirt, and set-in long sleeves with rib-knit wristlets (see Figure 1). The skirt and wristlets shall be an integral part of the body and sleeves respectively (not seamed or looped).

3.12.1 Figure. Figure 1 is furnished for information purposes only. When inconsistencies exist between the written specification and the figure, the written specification shall govern.

3.13 Patterns. The patterns for cutting all component parts of the sweater shall be furnished by the contractor, and shall be of the proper proportions to provide sweaters conforming to the finished measurements specified in 3.16.

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3.14 Construction.

3.14.1 Stitches, seams, and stitching. Stitch, seam, and stitching types specified in Table III shall conform to FED-STD-751. Whenever two or more stitch types are given for the same operation, any type may be used. Seam allowances shall be maintained with seams properly sewn so that no raw edges, runoffs, twists, pleats, or open seams result. The bight of all overedge stitched seams shall be 1/4 to 5/16 inch. The gauge of the type 402 and 406 cover stitching shall be 1/4 to 5/16 inch.

3.14.1.1 Cover and overedge stitching. When stitch type 402 or 406 is used, the yarn specified in 3.4.2 shall be for the cover stitching operation. When type 504 or 505 is used, the yarn specified in 3.4.2 shall be used for overedging.

3.14.2 Stitches per inch. The minimum and maximum number of stitches per inch shall be as specified in Table III.

3.14.3 Thread breaks and ends of seams. Ends of all seams and stitchings, when not caught in other seams or stitchings, shall be securely tacked or backstitched not less than 1/4 inch. Thread breaks (all stitch types) shall be secured by stitching back of the break not less than 1/2 inch.

3.15 Manufacturing operations requirements. The sweater shall be manufactured in accordance with operation requirements specified in Table III. The contractor is not required to follow the exact sequence of operations.

3.15.1 Shade marking. The component parts of the sweater shall be shade marked to insure a uniform shade and size throughout the garment. Any method of shade marking may be used except:

a. Corrosive metal fastening devices.
b. Sew-on shade tickets.
c. Adhesive type tickets which show discoloration or the adhesive mass adheres to the material upon removal of the tickets.

3.15.2 Abbreviations in table of operations. The abbreviations used in Table III are as follows:

Stch	-	Stitch
in	-	inch
Ndl	-	Needle
Bob	-	Bobbin
Lp	-	Looper
Cv	-	Cover
Brtck	-	Bartack

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TABLE III - CONSTRUCTION OF SWEATER

No.	Operation	Stch Type	Seam/ Stch type	Stch in	Thread Ndl Bob/ Lp/Cv
NOTE:	Caution should be taken to insure that the bight of all 504 and 505 stitching be maintained at 1/4 inch to 5/16 inch for the entire length of the seam.				
1/	Use the yarn specified in 3.4.2.				
1.	Cutting a. Cut the cloth component parts of the sweater in strict accordance with the contractor's patterns, which shall show size, shape, and directional lines for proper assembly and applicable matching of all parts.				
2.	Replacement of defective components a. During the spreading, cutting, and manufacturing process, components having material defects or damages that are classified as defects in Section 4 shall be removed from production and replaced with non-defective and properly matched components.				
3.	Shade and size marking (see 3.15.1) a. All component parts shall be marked, ticketed, or bundled to insure a uniform shade, size, and proper assembly throughout the sweater.				
4.	Join side seams - optional for body and skirt a. Join and close cut sides of the continuously knitted body and skirt to form two (2) full side seams.	504 or 505	SSa-1	8-10	A/3 A/3; 1/
	b. Turn under stitching chain at bottom edge of each side seam, and position a 1/4 to 1/2 inch horizontal bartack across seam 1/16 to 3/16 inch from finished bottom edge of skirt, or 2 1/2 to 3 inches of stitching chain may be buried in the closing seam by back latch tacking, using a chain off thread back tacking attachment JUKI-Z015 or equivalent.	Brtck	-	21-28 per brtck	A/3 A/3;

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TABLE III - CONSTRUCTION OF SWEATER

No.	Operation	Stch Type	Seam/ Stch type	Stch in	Thread Ndl	Thread Bob/ Lp/Cv
5.	Join shoulders					
	a. Join fronts to back along shoulder edges.	504 or 505	SSh-2(a)	8-10	A/3	A/3; 1/
	b. Cover stitch each seam.	402 or 406	SSh-2(b)	12-14	A/3	1/
6.	Attach collarette					
	a. Circular knit seamless tubing, when applicable: 1. Cut and fold the collarette to finish in the length (wale) direction.					
	2. Join and close raw edges of the collarette to raw edge of sweater neckline opening.	504 or 505	SSh-2(a)	8-10	A/3	A/3; 1/
	3. Cover stitch seam.	402 or 406	SSh-2(b)	12-14	A/3	1/
	OR					
	b. Flat knit collarette piece, when applicable: 1. Cut the material to finish in the length (wale) direction.					
	2. Join and close raw edges of the material to form collarette.	504 or 505	SSa-1	8-10	A/3	A/3; 1/
	3. Fold to finish with the merrow (overedge) stitch on the inside and raw edges along the bottom.					
	4. Join and close raw edges of collarette material to the raw edge of sweater neckline opening. The collarette seam of operation 6b2 should finish aligned (± 1/4 inch) with one of the shoulder seams.	504 or 505	SSh-2(a)	8-10	A/3	A/3; 1/
	5. Cover stitch seam.	402 or 406	SSh-2(b)	12-14	A/3	1/

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TABLE III - CONSTRUCTION OF SWEATER

No.	Operation	Stch Type	Seam/ Stch type	Stch in	Thread Ndl	Thread Bob/ Lp/Cv
	c. Regardless of the applicable knitting process and size sweater being procured, the collarette, when measured on a straight line at the center from the top folded edge to bottom edge at sweater neckline, shall finish 2 to 2 1/2 inches in length (wale) direction.					
7.	Close sleeves					
	a. Close sleeves and wristlets at inseams.	504 or 505	SSa-1	8-10	A/3	A/3; 1/
	b. Turn under stitching chain at bottom edge of each wristlet inseam, and position a 1/4 to 1/2 inch horizontal bartack across seam 1/16 to 3/16 inch from finished bottom edge of the wristlet or 2 1/2 to 3 inches of stitching chain may be buried in the closing seam by back latch tacking using a chain off thread back-tacking attachment JUKI-Z015 or equivalent.	Brtck	-	21-28 per brtck	A/3	A/3
8.	Attach sleeves to armholes					
	a. Set and stitch sleeves to armholes.	504 or 505	SSh-2(a)	8-10	A/3	A/3 1/
	b. Cover stitch each seam.	402 or 406	SSh-2(b)	12-14	A/3	1/
9.	Attach label (see 3.4.5)					
	a. Position the label on inside of sweater at center back, 1/4 to 3/4 inch below collarette seam. Stitch the label vertically along both ends, 1/8 to 1/4 inch from the edges, keeping label slack so that the material on the underside may stretch fully when worn.	301		12-14	A/3	A/3
10.	Cleaning					
	a. All ends of thread shall be trimmed and all loose threads removed throughout the sweater. Care shall be taken so that damage to the sweater does not result.					
	b. Remove all spots and stains.					
	c. Remove all shade tickets.					

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3.16 Measurements. The sizes and measurements of the finished sweater shal conform to the dimensions specified in Table IV. All measurements ar expressed in inches, and shall be made with the sweater in a fully relaxe state, placed smooth and flat.

Table IV - Finished measurements

Measurement	Sizes					
	Extra Small (32-34)	Small (36-38)	Medium (40-42)	Large (44-46)	Extra Large (48-50)	Tolerances
Neck width (A)	7 1/2	7 1/2	8	8	9	± 1/2
Chest (B)	15	17	19	21	23	± 1
Length (C)	23 1/2	24 1/2	25 1/2	27	28 1/2	± 1
Overarm sleeve length (D)	20 1/2	21	22	23 1/2	24 1/2	± 1
Armhole length (E)	7 1/2	7 3/4	8	8 1/2	9 1/4	± 1/2
Wristlet width (F)	3	3	3	3 1/2	3 1/2	± 1/2
Sleeve width (G)	5 1/2	5 3/4	6	6 1/2	7	± 1/2

NOTE: A through G refer to Figure 1.

A. Measurement shall be taken from the intersection of the collarette seam an shoulder seam across neck front to similar intersection on opposite side.
B. Measurement shall be taken across front approximately 1 inch below armhol seam, folded edge to edge. When two full side seams are used, measure fro seam to seam.
C. Measurement shall be taken from the intersection of the collarette seam an shoulder seam perpendicular to bottom edge of skirt.
D. Measurement shall be taken along outside folded edge of sleeve from th intersection of the armhole seam and shoulder seam to bottom edge o wristlet.
E. Measurement shall be taken from the intersection of the armhole seam an shoulder seam on a straight line to the intersection of the armhole sea and sleeve inseam.
F. Measurement shall be taken at center of wristlet on a straight line, folde edge to seamed edge.
G. Measure across width of sleeve from the folded edge to the sleeve seam at point 12 inches below the shoulder.

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3.17 Workmanship. The finished sweater shall conform to the quality of product established by this specification. The occurence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article inspection. The first article, submitted in accordance with 3.3, shall be inspected for compliance with design, construction, workmanship, and dimensional requirements.

4.4 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated.

4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable procurement documents. A certificate of compliance shall be submitted for the modacrylic fiber requirements of 3.4.1.1 and the wool fiber requirements of 3.4.1.2 (except for wool grade).

4.4.1.1 Wool grade. The wool shall be visually examined for grade in top form by comparison with the applicable US Standard (see 2.2). In the event of a dispute resulting from this examination, the wool grade shall be determined by the width (wedge) method, approved by the government. The sample unit shall be one yard of undyed top. The sample size shall be as shown in Table V. The lot size shall be expressed in units of pounds, and shall be unacceptable if a single unit of product fails to meet the required grade.

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Table V - Sample size

Lot size	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

4.4.2 Examination of the end item. Examination of the end item shall be in accordance with 4.4.2.1 through 4.4.2.2. The applicable inspection levels and acceptable quality levels (AQL's) shall be as indicated in 4.4.2.3. The lot size shall be expressed in units of sweaters. The sample unit shall be one finished sweater.

4.4.2.1 Visual examination.

Defect	Classification Major	Minor
Material Defects and Damages		
a. Any hole, cut, tear, smash, burn, needle chew, missing yarn, dropstitch, broken yarn	X	
b. Any knot or untrimmed end of knot on outside		X
c. Cockled fabric, i.e., evidence of distorted, wavy or ridgy area exceeding 3 inches in the longest dimension	X	
d. Heavy yarn or slub more than twice the diameter of the yarn, or thin place, measuring more than 1/2 inch in the longest direction on the outside		X
e. Color not as specified	X	
NOTE: For knitted fabric defect definitions, see MIL-STD-1491.		
Cleanness		
a. Spot or stain on outside (measured in longest direction)		
- up to 1/4 inch, inclusive		X
- more than 1/4 inch	X	
b. Three or more threads or yarns on outside and inside loose, thread ends on inside not trimmed, or not removed		X
c. Any marking exposed or visible on outside		X
d. Two or more shade or size tickets not removed		X
Shaded parts		
a. Any part shaded or variation in shade within a part		
1. On outside	X	

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Defect	Classification Major	Minor
Components and Assembly		
a. Any component part or operation omitted (unless otherwise classified herein)	X	
b. Any component not as specified, i.e., type, color, etc.	X	
c. Any component part tight, twisted, distorted, pleated, or full (unless otherwise classified herein)		X
d. Any operation not as specified (unless otherwise classified herein)		X
Seams and Stitching		
a. Accuracy of seaming:		
1. Seam puckered		X
2. Seam irregular, twisted, pleated, or wavy		X
3. Any part of sweater caught in an unrelated operation	X	
4. Ends of seams or stitching, when not caught in other seams or stitching, backtacked less than 1/4 inch		X
5. Thread breaks not secured or stitching back of the break less than 1/2 inch.		
b. Gage of stitching:		
1. For all stitch types other than 504 or 505: Irregular, not within range specified, or varies more than 1/16 inch when no range is specified (score only when the condition exists on more than 1/2 the length of seam)		X
2. For 504 or 505 stitch types: Irregular, not within range specified (score even when condition exists for less than 1/2 the length of the seam)	X	
c. Seam and stitch type:		
1. Not specified seam or stitch type		X
2. Any line of stitching omitted	X	
3. Any line of stitching misplaced or not beginning or ending where specified		X
d. Any open seam	X	

NOTE: The seam shall be classified open when one or more stitches joining a seam are broken or when two or more continuous skipped or runoff stitches occur.

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Defect	Classification Major	Minor
e. Stitch tension		
1. Loose tension resulting in a loose seam	X	
2. Tight tension (stitches break when normal strain is applied in the direction of the seam or stitching)	X	
f. Stitches per inch more or less than specified		X
g. Bartack or backtacking		
1. Any omitted, insecure, misplaced or not serving intended purpose		X
2. Length less than 1/4 inch		X
a. Bartack		
b. Backtacking: Amount of stitching chain buried in closing seam less than 2-1/2 inches or more than 3 inches		X
Label		
a. Omitted, incorrect, or illegible	X	
b. Misplaced, not attached as specified, or stitching through printing		X

4.4.2.2 Dimensional examination. The sweater shall be examined for defects in dimensions in accordance with the following:

a. Any measurement deviating from nominal dimensions and tolerances specified in Table IV shall be scored as a size measurement defect.
b. Sleeve lengths unequal in length by 1/2 inch or more shall be scored as a size measurement defect.
c. Finished height (wale) direction of skirt less than 3 1/4 inches or more than 3 3/4 inches shall be scored as a size measurement defect.
d. Finished height (wale) direction of wristlet less than 3 1/4 inches or more than 3 3/4 inches shall be scored as a size measurement defect.
e. Finished height of collarette (wale) direction less than 2 inches or more than 2 1/2 inches shall be scored as a size measurement defect (see Table III, operation 6c).
f. Finished sleeve not shaped as shown in Figure 1.

4.4.2.3 Inspection levels and acceptable quality levels (AQL's). The inspection levels and the acceptable quality levels, expressed in defects per 100 units for visual and dimensional examination, shall be as follows:

	Inspection level	AQL's Major	Total
For defects applicable to 4.4.2.1	I	2.5	6.5
For defects applicable to 4.4.2.2	S-3	-	4.0

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4.4.3 Examination of packaging requirements. An examination shall be made to determine that packaging, packing, and marking comply with the Section 5 requirements of this specification. The sample unit shall be one shipping container fully prepared for delivery except that it need not be closed. Examination for the closure defects listed below shall be made. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5 defects per 100 units.

Examine	Defect
Marking (exterior and interior)	Omitted, incorrect, illegible, of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, improper taping, loose strapping, inadequate stapling, bulging or distortion of container, open and non-continuous heat sealed seams of polyethlyene bag, or incorrectly fabricated bag.
Content	Number of sweaters per shipping container is more or less than specified 1/

1/ For this defect, one shipping container in the sample shall be examined.

4.4.4 Palletization examination. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully packaged. The lot size shall be the number of palletized unit loads in the end item inspection lot. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5 as specified in MIL-STD-105.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirements.
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.

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4.5 Testing of the end item. The methods of testing specified in FED-STD-191, whenever applicable, and as listed in Table VI, shall be followed. The physical and chemical values specified in Section 3 apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test methods. The lot size shall be expressed in units of one (1) sweater each. With the exception of weight per dozen, the sample size shall be as specified in Table V. The sample unit shall be as follows:

For weight - Three (3) sweaters of each size shown in the lot regardless of lot size.
All other tests - Two (2) sweaters

The lot shall be unacceptable if one or more sample units fail to meet any of the test requirements specified.

Table VI - End item tests

Characteristics	Requirement Paragraph	Test Method
Fiber content	3.4.2	2102
Type of knit stitch:		
All component parts	3.4.3	Visual 1/
Wales per inch:		
All component parts	3.4.3	5070
Courses per inch:		
All component parts	3.4.3	5070
Colorfastness to:		
Light	3.5.2	5660
Laundering	3.5.2	5614
Wet-dry cleaning	3.5.2	5622
Perspiration	3.5.2	5680
Crocking	3.5.2	5651
Nonfibrous material content	3.6	2611
Shrinkage	3.7	AATCC-135-1978 2/
Bursting strength	3.8	5120
Flame resistance		
After flametime, sec.	3.9	5903
After glow time, sec.	3.9	5903
Char length, inch	3.9	5903
pH	3.10	2811
Weight per dozen	3.11	3/

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1/ One determination shall be performed in each area of the sample unit, and the results reported as "pass" or "fail".

2/ Testing shall be performed under the following conditions: Machine cycle #2 (delicate), wash temperature II (120°F), drying procedure B (tumble dry) 5 cycles.

3/ The three (3) sweaters of each size encompassing the sample unit for testing, shall be allowed to reach equilibrium under standard conditions as defined in FED-STD-191. When under standard conditions, the sweaters shall be weighed as a unit to the nearest 1.0 ounce, and that result multiplied by 4 to obtain the weight per dozen. The entire lot of the applicable size shall be rejected if the weight per dozen is below the minimum weight specified.

5. PACKAGING

5.1 Preservation-packaging. Preservation packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Each sweater shall be neatly folded to measure approximately 14-1/2 inches in length by 11-1/2 inches in width. Each folded sweater face side up shall be inserted into a snug-fitting, flat style, clear polyethlene film bag of 0.0015 inch (0.04mm) thickness (± 25 percent tolerance). The bag may be fabricated from polyethylene film tubing or sheeting, and shall be not less than 1 inch greater in length than the folded garment. The polyethylene bag shall be formed with heat sealed seams that are straight, continuous, and parallel to each other and the formed edges of the bag. The final closure of the bag shall be heat sealed, with the heat seal made as close as possible to the open end. A 1/4 inch diameter hole shall be made at one corner of the bag to allow excess air to escape. Alternatively, the final closure of the polyethylene bag may be accomplished by means of a tuck or reverse flap.

5.1.2 Level C. Sweaters shall be preserved-packaged to afford adequate protection against physical damage during shipment from the contractor to the first receiving activity. The package and the quantity per package shall be the same as that normally used by the contractor for retail distribution.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Twenty-six (26) sweaters, of one size only and packaged as specified in 5.1, shall be packed in a corrugated fiberboard shipping container conforming to type CF, class weather-resistant, variety DW, grade V15c, size 3A of MIL-B-17757. The fiberboard for the box liner shall conform to type CF, class domestic of MIL-B-17757. Level A packages shall be packed flat, two in length, one in width, and thirteen in depth within the shipping container. Each container shall have the contents completely covered on the top and bottom with a sheet of commercial grade kraft paper. Toward the end of the contract, or when there are less than the required amount per container of the same size, mixed sizes may be packed within the same shipping container.

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5.2.2 Level B. Twenty-six (26) sweaters, of one size only and packaged as specified in 5.1, shall be packed in a corrugated fiberboard shipping container conforming to type CF, class domestic, variety DW, grade 275, size 3A, of MIL-B-17757. The fiberboard for the box liner shall conform to type CF, class domestic of MIL-B-17757. Level A packages shall be packed flat two in length, one in width, and thirteen in depth within the shipping container. Each container shall have the contents completely covered on the top and bottom with a sheet of commercial grade kraft paper. Toward the end of the contract, or when there are less than the required amount per container of the same size, mixed sizes may be packed within the same shipping container.

5.2.3 Level C. Sweaters, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with the US Postal Service Manual, Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Palletization. When specified (see 6.2), sweaters packed as specified in 5.2, shall be palletized on a 4-way entry pallet in accordance with load type IA of MIL-STD-147. Each prepared load shall be bonded with primary and secndary straps in accordance with the bonding means (C,K and L) or O or P. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. The pallet shall be 4-way, Type III, Class 1, Style A, Size 2, wood group I, Grade A of NN-P-71, or 4-way, Style (1, 1A or 1B), Size A, Type I Class 1 of MIL-P-15011. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

5.4 Marking. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

5.4.1 Polyethylene bagged packages. Polyethylene bagged packages shall have the following information legibly printed or stamped in black directly on the bag across the center face or on a white paper label inserted within the bag so as to permit ready identification:

SWEATER, MAN'S (FLAME RETARDANT)
STOCK NO.:
SIZE:
QUANTITY:

5.4.2 Labels, mixed sizes. Each shipping container, packed with mixed sizes, shall have securely attached to the end and side directly under the printing or stenciling, a white paper label 5 by 4 inches with the words "MIXED SIZES" plainly printed or stamped thereon and under these words shall be legibly printed or stamped the correct quantity and sizes contained therein.

6. NOTES

6.1 Intended use. The sweater covered by this specification is intended for use by male military personnel of the Department of the Navy as part of the utility uniform.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Sizes required (see 1.2).
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
d. When first article inspection is required (see 3.3), the item will be tested and should be a first article sample. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, quantity, and testing and approval.
e. When palletization is required (see 5.3).

6.3 Samples. For access to samples, address the procuring office issuing the invitation for bids.

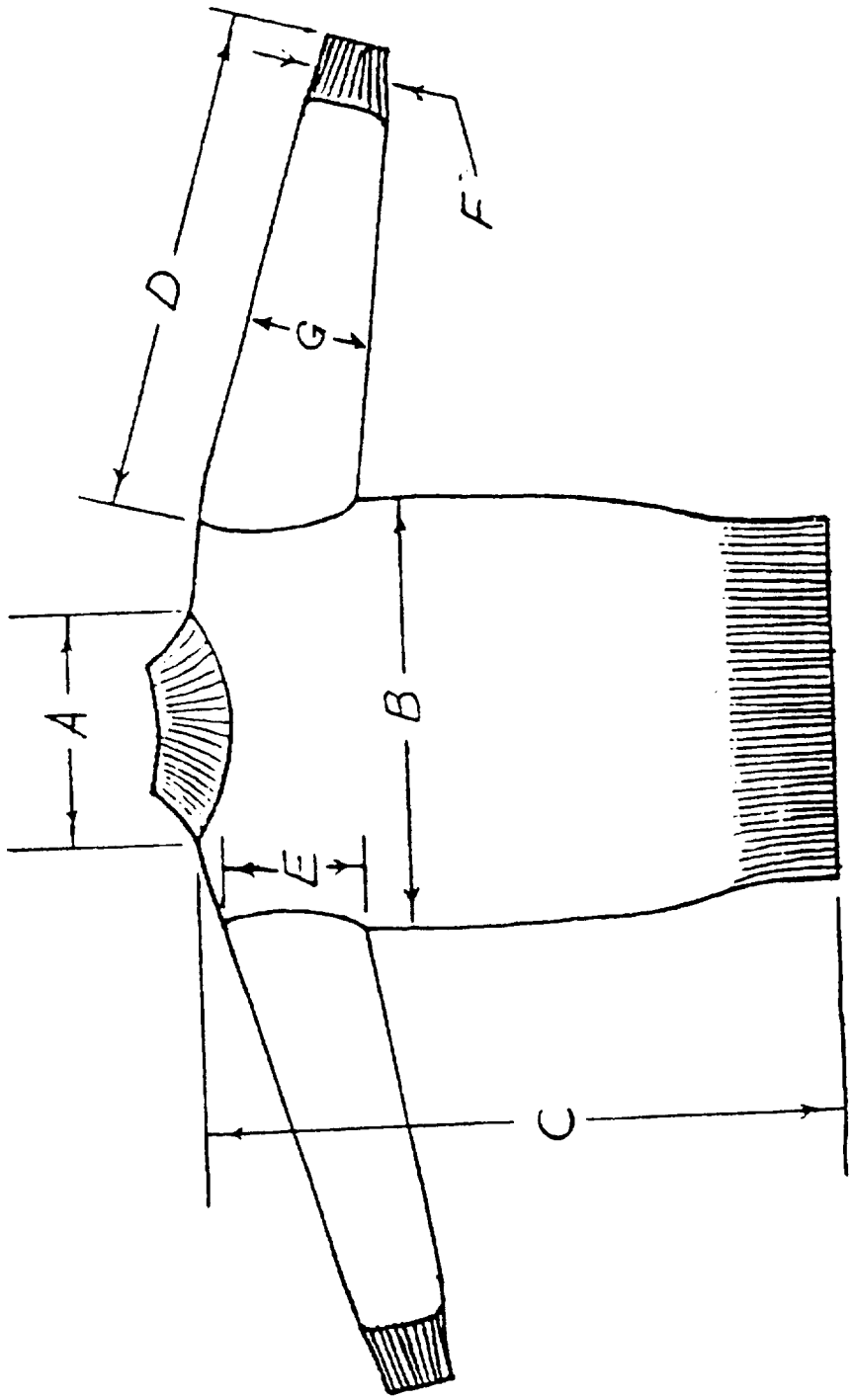
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SWEATER, MAN'S (FLAME RETARDANT)

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(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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